

Combining Generative and Discriminative Models for Hybrid Inference

Intelligent systems
Bayesian Multi Modeling

2025

Motivation: Two Approaches to Inference

Generative Models (Graphical Models)

- ▶ Encode prior knowledge
- ▶ Strong inductive bias
- ▶ Excel in low-data regime
- ▶ May miss real-world complexity

Discriminative Models (Neural Networks)

- ▶ Learn directly from data
- ▶ High model capacity
- ▶ Require large datasets
- ▶ May lack interpretability

Key Question: Can we combine both for better performance across all data regimes?

Core Idea: Hybrid Inference

Combine:

- ▶ **GM-messages:** Derived from graphical model (e.g., physics equations)
- ▶ **GNN-messages:** Learned from data to correct GM-messages

The Hybrid Update Rule:

$$x_k^{(i+1)} = x_k^{(i)} + \gamma \left(M_k^{(i)} + \epsilon_k^{(i+1)} \right)$$

where:

- ▶ $M_k^{(i)}$: Graphical model messages (traditional inference)
- ▶ $\epsilon_k^{(i+1)}$: Learned correction signal (residual learning)

Benefits:

- ▶ Uses inductive bias when data is scarce
- ▶ Leverages neural network capacity when data is abundant
- ▶ Gracefully interpolates between regimes

Hidden Markov Process (HMM) Framework

Problem Setup:

- ▶ Hidden states: $x = \{x_1, \dots, x_K\}$
- ▶ Observations: $y = \{y_1, \dots, y_K\}$
- ▶ Goal: Infer states $\hat{x} = \arg \max_x p(x|y)$

Graphical Model Factorization:

$$p(x, y) = p(x_0) \prod_{k=1}^K p(x_k | x_{k-1}) p(y_k | x_k)$$

Kalman Filter (Linear Case):

$$\begin{aligned} x_k &= Fx_{k-1} + q_k, & q_k &\sim \mathcal{N}(0, Q) \\ y_k &= Hx_k + r_k, & r_k &\sim \mathcal{N}(0, R) \end{aligned}$$

Limitation: Real-world systems are non-linear and complex!

Graphical Model Messages (GM-Messages)

Iterative Optimization Formulation:

$$x_k^{(i+1)} = x_k^{(i)} + \gamma M_k^{(i)}$$

where $M_k^{(i)}$ combines three directional messages:

$$M_k^{(i)} = \mu_{x_{k-1} \rightarrow x_k}^{(i)} + \mu_{x_{k+1} \rightarrow x_k}^{(i)} + \mu_{y_k \rightarrow x_k}^{(i)}$$

For Linear Gaussian Case (Kalman Filter):

$$\mu_{x_{k-1} \rightarrow x_k} = -Q^{-1}(x_k - Fx_{k-1})$$

$$\mu_{x_{k+1} \rightarrow x_k} = F^T Q^{-1}(x_{k+1} - Fx_k)$$

$$\mu_{y_k \rightarrow x_k} = H^T R^{-1}(y_k - Hx_k)$$

Advantage: Messages encode physical/structural knowledge!

Graph Neural Network Messages (GNN-Messages)

GNN message passing on equivalent graph:

1. **Node representation:** $h_x^{(0)}$ (random init), $h_y^{(0)}$ (from y_k via linear layer)

2. **Edge message computation:**

$$m_{k,n}^{(i)} = z_{k,n} f_e(h_{x_k}^{(i)}, h_{v_n}^{(i)}, \mu_{v_n \rightarrow x_k})$$

3. **Aggregate messages to node:**

$$U_k^{(i)} = \sum_{v_n \neq x_k} m_{k,n}^{(i)}$$

4. **GRU update:**

$$h_{x_k}^{(i+1)} = \text{GRU}(U_k^{(i)}, h_{x_k}^{(i)})$$

5. **Compute correction factor:**

$$\epsilon_k^{(i+1)} = f_{\text{dec}}(h_{x_k}^{(i+1)})$$

Key idea: Learn to model **residual errors**, not full dynamics!

Training Procedure

Three-step training:

1. **Initialization:** Set $x_k^{(0)}$ to maximize $p(y_k|x_k)$
 - ▶ Example: position = observed position
2. **Tune graphical model:** Optimize variances in Gaussian distributions
 - ▶ Treat as standard Kalman filter calibration
3. **Train GNN component:** Minimize weighted loss

$$\text{Loss}(\Theta) = \sum_{i=1}^N w_i L(y_{\text{true}}, \hat{x}^{(i)})$$

where $w_i = i/N$ (emphasize later iterations)

Hyperparameters:

- ▶ Learning rate: 10^{-3} (Adam optimizer)
- ▶ Step size: $\gamma = 0.005$
- ▶ Iterations: $N = 50$
- ▶ Hidden features: $n_f = 48$

Experiment 1: Linear Dynamics

Setup: Synthetic trajectories with known non-uniform acceleration and air drag

$$\begin{aligned}\frac{\partial p}{\partial t} &= v \\ \frac{\partial v}{\partial t} &= a - cv \\ \frac{\partial a}{\partial t} &= -\tau v\end{aligned}$$

Graphical model: Assumes **uniform motion only** (missing physics!)

Results:

- ▶ **Low data (<1K):** Kalman Smoother wins (model is approximately correct)
- ▶ **Medium data (1K-10K):** Hybrid dominates (leverages both)
- ▶ **High data (>10K):** Hybrid and GNN compete, Hybrid edges ahead

Experiment 1: Results (Linear Dynamics)

Mean Squared Error vs. Number of Training Samples:

Method	100 samples	1K samples	100K samples
Kalman Smoother	Best	Better	Worst
GNN only	Worst	Good	Best
Hybrid	Best	Best	Best

Key Insight: Hybrid model achieves superior performance across all data regimes!

Experiment 2: Lorenz Attractor (Chaotic System)

System: Non-linear chaotic dynamics (atmospheric convection)

$$\begin{pmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{pmatrix} = \begin{pmatrix} -10(z_2 - z_1) \\ z_1(28 - z_3) - z_2 \\ z_1 z_2 - \frac{8}{3} z_3 \end{pmatrix}$$

Graphical model approximation: Taylor expansion (1st, 2nd, or 5th order)

Key Challenge: Transition matrix F_{x_k} depends on state—very hard to learn from scratch!

Sample Efficiency Results:

Method	MSE = 0.1	MSE = 0.05
GNN (J=0)	~ 5,000 samples	~ 90,000 samples
Hybrid (J=1)	~ 500 samples	~ 5,000 samples
Hybrid (J=2 or 5)	~ 400 samples	~ 4,000 samples

Result: Hybrid achieves **10–20× sample efficiency** over pure GNN!

Experiment 2: Lorenz Results Visualization

Qualitative Comparison (5K-length test trajectory):

Method	Mean Squared Error
Observations (baseline)	0.2462
GNN only	0.0613
E-Kalman Smoother	0.0372
Hybrid Model	0.0169

Observation: Even with non-linear approximation, hybrid model **outperforms both** GNN and E-Kalman Smoother!

Why? Residual dynamics (inference error) are more linear than original dynamics—easier for GNN to learn!

Experiment 3: Real-World Data (Michigan NCLT Robot)

Dataset: GPS-equipped robot (Segway) traversing University of Michigan campus

- ▶ Noisy GPS measurements (y)
- ▶ Ground truth position (x)
- ▶ Goal: Denoise GPS, estimate true position

Graphical Model: Uniform motion assumption (overly simplistic)

Results (MSE):

Method	MSE
Raw observations	3.4974
Kalman Smoother	3.0099
GM-messages only	3.0048
GNN only	1.7929
Hybrid Model	1.4109

Conclusion: Hybrid model achieves **53% improvement** over Kalman Filter on real-world data!

Why Does Hybrid Work? (The Residual Learning Insight)

Core principle: Residuals are easier to learn than full signals

Traditional GNN approach:

$$\hat{x}_k = f_{\text{GNN}}(y)$$

Learn full dynamics from scratch—hard!

Hybrid approach:

$$\hat{x}_k = \underbrace{x_k^{\text{GM}}}_{\text{structured prior}} + \underbrace{\epsilon_k}_{\text{learned correction}}$$

Learn only the “mistakes” of the graphical model—easier!

Evidence:

- ▶ Lorenz: GNN struggles even with 100K samples
- ▶ Hybrid + weak physics ($J=1$): converges with 500 samples
- ▶ Suggests residual errors have simpler structure

Relationship to Related Work

Hybrid inference connects multiple research areas:

- ▶ **Meta-learning:** Learning flexible messages on top of algorithms
- ▶ **Structured prediction:** Exploiting output structure in discriminative models
- ▶ **Recurrent inference machines:** Embedding generative models in RNNs
- ▶ **Graph neural networks for inference:** Learning message passing on graphical structures
- ▶ **Hybrid learning methods:** Combining symbolic and neural approaches

Novel contribution: Unlike prior work, hybrid model **runs both GM and GNN in parallel**, allowing automatic balancing via cross-validation

Generalizability: Beyond HMMs

Current scope: Sequential models (Hidden Markov Models)

General framework: Can be applied to arbitrary graphical structures!

- ▶ Equations derived on **edges and nodes**
- ▶ By modifying input graph structure, apply to:
 - ▶ Undirected models (Markov random fields)
 - ▶ General factor graphs
 - ▶ Discrete variable models

Future directions:

- ▶ Learn graphical model structure itself
- ▶ Bayesian treatment of prior/data tradeoff (replacing cross-validation)
- ▶ Improved interpretability through graphical model backbone
- ▶ Applications to other inference problems

Summary and Key Takeaways

Main Contribution:

- ▶ Hybrid algorithm combining **graphical inference** + **learned inference**
- ▶ Automatically balances using cross-validation
- ▶ Unifies message passing with graph neural networks

Empirical Results:

- ▶ Linear dynamics: Outperforms both methods across all data regimes
- ▶ Lorenz attractor: 10–20 $\times$ sample efficiency gain over GNN
- ▶ Real-world GPS: 53% improvement over Kalman filter

Key Insight:

- ▶ Learning residual corrections is easier than learning full dynamics
- ▶ Inductive bias from graphical models remains valuable in big-data era
- ▶ Graceful interpolation between small and large data regimes

Discussion and Questions

Strengths:

- ▶ Well-motivated problem with clear practical benefits
- ▶ Principled hybrid approach grounded in message passing
- ▶ Comprehensive experiments on synthetic and real data
- ▶ Elegant residual learning formulation

Limitations and Future Work:

- ▶ Cross-validation for balancing; Bayesian approach could be more principled
- ▶ Limited to sequential models in current work
- ▶ Interpretability claims should be further developed
- ▶ Computational cost analysis not provided

Thank you!

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